

Ibrahim Al Omran

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Education

McMaster University

B.Eng, Software Engineering

Expected 2027

Hamilton, ON

Technical Skills

Languages: C, Java, JavaScript, Python, Verilog, Bash, HTML/CSS, TypeScript

Technologies: React, Node.js, Firebase, Prisma, Supabase, Tailwind CSS, Git, FastAPI, Next.js

Team Projects

CDL Career Mode (Next.js, Supabase, FastAPI) - *Team Project*

App

- Developed an esports management simulator modeling Call of Duty League team dynamics, player ratings, and contract markets.
- Implemented Elo-based progression and **mode-specific telemetry** (Hardpoint, Control, S&D) to generate realistic player stats.
- Engineered Supabase schema for real-time simulation storage, reducing cold-start latency by 75% and enabling thousands of dynamic stat updates per run.

Resume Matcher (Next.js, FastAPI, LLaMA 3) - *Team Project*

App — Repo

- Created an AI-powered resume optimizer that matches resume to job postings and auto-generates tailored **LaTeX** edits.
- Utilized **Groq LLaMA 3** + **FastAPI** for contextual analysis, improving match quality by 40% vs. keyword-based models.
- Integrated PDF parsing and **LaTeX** compilation for seamless resume regeneration.

Rescue Drone Mission (Java) - *McMaster University*

Repo

- Created a simulated drone exploration system for islands with safe navigation and return-home logic.
- Applied robust **OOP design patterns** and error handling; contributed 30% of code in 3-member team.
- Practiced **Agile methodologies** using Kanban boards to coordinate tasks and ensure iterative progress.

Personal Projects

Schedules (TypeScript, Supabase) - *Personal Project*

App — Repo

- Built an app that parses **XLSX** schedules to display shifts, coworkers, and sync with **Google Calendar**.
- Adopted **Prisma** and **Supabase** for reliable storage; used actively by 15+ coworkers at Crocs Inc.
- Enhanced workplace productivity by delivering accurate, automated shift visualization, reducing scheduling time by 90%.

PinPoint Geography Quiz (React, Firebase) - *Personal Project*

App — Repo

- Developed an interactive geography quiz with survival mode and global leaderboard (15+ users).
- Implemented real-time tracking with **Firebase** and deployed globally on **Vercel**.
- Crafted an engaging gameplay loop with live leaderboards and competitive streak tracking.

Autonomous Recycling System (Python) - *McMaster University*

Repo

- Programmed a bottle-sorting system using weight/color sensors and robotic arms in **QLabs** simulation.
- Filtered noisy input to achieve 90%+ classification accuracy across 50+ rigorous tests.
- Designed robust sensor-control pipelines that elevated system precision and reliability.

Dynamic Tic Tac Toe (Java) - *McMaster University*

Repo

- Engineered a scalable Tic Tac Toe with variable board sizes and dynamic win conditions.
- Implemented modular **OOP** game logic and a responsive interactive UI.
- Delivered gameplay flexibility through an adaptive rules engine for diverse board configurations.

Awards

McMaster Award of Excellence

Highest Average in Grade 11 and 12 Mathematics